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**"Unlocking Customer Loyalty with Voucher Possession: Influence of Points
Accumulation on Purchase Behavior of Loyalty Program Members"**

By

Ntontis Spiros | S5124999

University of Groningen

Master Marketing Analytics and Data Science (MSc)

Faculty of Economics & Business – Department of Marketing

1st Supervisor: Dr. Stephanie Vigier-Zouhar

2nd Supervisor: Prof. dr. T.H.A. Bijmolt

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ABSTRACT

Over the years, loyalty programs have gained increasing significance across various industries. As managers and retailers strive to implement effective marketing strategies for customer attraction and retention within these programs, this study explores the influence of possessing a voucher on customer purchase behavior. The research investigates the relationship between the dependent variable, the number of items customers purchase “Item Quantity” and the independent variable “Reward Value”, calculated as the difference between the original price and the discounted price, representing the value of each customer's voucher. The independent variable is analyzed alongside the moderating effects of “Voucher Used” and “Age”. Focusing primarily on customers with a voucher, a binary variable, “Has Voucher,” serves as a control variable to indicate voucher possession. Customers without vouchers were excluded from subsequent analyses. The dataset is sourced from the reputable retail company operating within the Netherlands market. Given that the dependent variable is a count variable with a mean higher than the variance and due to the dependent variable, which is by definition larger than 0, Poisson Regression and Zero Truncated Poisson are utilized to explore three hypotheses. Contrary to Hypothesis 1, which suggested a positive impact of reward value on loyalty program members' purchase behavior, the results reveal a decrease in item quantity with higher reward values. Hypothesis 2, proposing that higher reward values enhance the purchase behavior of loyalty program members using vouchers, is rejected due to a lack of statistical significance. Similarly, Hypothesis 3, suggesting a higher reward value would increase the purchase behavior of older individuals, is rejected, with the results revealing the opposite effect. Recommendations include offering lower-value rewards to increase purchase behavior among loyalty program members and directing high-value rewards towards younger customers. The study also identifies uncertainties associated with voucher redemption challenges. The conclusion highlights the study's limitations and underscores the importance of future research, providing suggestions for further exploration in loyalty programs.

Keywords: Loyalty Programs, Point Accumulation, Reward Value, Item Quantity, Voucher Used, Age, Poisson Regression, Zero Truncated Poisson

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1. INTRODUCTION

Understanding the dynamics of customer loyalty is particularly crucial in the context of loyalty programs (LPs), designed to ensure repeated engagement over time. In recent times, loyalty programs have gained significance across various industries, (Leenheer and Bijmolt, 2003), and their primary objective is not just to attract customers but to enhance customer retention, especially within valuable customer segments (Bolton, Kannan, and Bramlett, 2000). Crucial to the success of LPs is the role of rewards, which serves as a catalyst in shaping the behavior of loyalty program members. As a result, loyalty programs have become integral to businesses' customer relationship management initiatives, with diverse approaches in their implementation (Kumar and Reinartz, 2006; Shugan, 2005; Yi and Jeon, 2003).

This thesis deeply delves into the domain of rewards within loyalty programs (LPs), with a particular focus on the intricacies of points accumulation. It has been observed that many organizations have made reward points their major marketing tool throughout time (Kivetz & Simonson, 2002; Nunes & Dreze, 2006; Uncles, Dowling, & Hammond, 2003), because consumers tend to increase their purchase rate as they approach the required number of points to earn a reward, (Kivetz, Urminsky, and Zheng, 2006).

A well-known and sufficient reward of points accumulation is the method that loyalty program members earn points by making purchases from the retailer, for example with every ... points (equivalent to ..€ spent) resulting in a ..€ loyalty voucher. As Hsee, Yu, Zhang & Zhang (2003) have noted that loyalty program points are awarded to customers after they make purchases from a company and can be saved and spent much like conventional money, even though they are not in the form of cash.

This reward mechanism aims to establish a sense of reciprocity between the customer and the company, encouraging customers to engage in more business transactions, leading to further rewards offered by the company, and so on.

This research aims to thoroughly analyze the online purchasing behaviors exhibited by participants in loyalty programs, according to the possession of the reward. The primary objective is to conduct an in-depth investigation into the factors influencing customers

online purchasing behavior, specifically examining the impact of variables such as the reward value, which indicates the amount of the reward, with the interaction term of the age of customers and the voucher utilization which depicts members who have acquired and used the voucher.

By delving into the analysis, it is crucial for managers, as they can gain actionable insights into optimizing their loyalty programs and increase the purchase rate of their customers. Subsequently, the main goal of this research is to examine and analyze if the possession of point accumulation significant wield over the purchase behavior of online loyalty program members.

Therefore, the following Research Question is derived:

“To what extent does the possession of Points Accumulation influence the purchase behavior of online loyalty program members?”

Examining the impact of points accumulation on the purchasing behaviors of online loyalty program members is highly relevant for businesses managing loyalty programs. The insights derived from this study offer practical guidance to managers. Specifically, the findings suggest optimization opportunities, including the consideration of lower reward values to boost purchase behavior among program members. Furthermore, it is suggested offering high-value vouchers to younger segments. While the results of voucher usage provide some lack of statistically significance.

In terms of scientific relevance, it is noteworthy that none of the hypotheses aligned with the expectations set by the literature review. Results revealed either contrary effects to the hypothesized outcomes or lacked statistical significance. This divergence emphasizes the complexity of customer behavior within loyalty programs and challenges established assumptions. By uncovering these unexpected patterns, the study contributes significantly to the scientific discourse, prompting a reconsideration of prevailing theories and encouraging further exploration.

This study is organized into several chapters. The literature review offers insights shaping the conceptual model, while the methodology outlines the research structure. The results chapter provides foundational analyses for the discussion, where findings are explored. The discussion synthesizes key findings and implications, followed by limitations and recommendations for future research.

2. LITERATURE OVERVIEW

2.1 Customer Loyalty

Customer loyalty, as defined by Baran, Galka, and Strunk (2008), is the strong bond that connects consumers to a company, retailer, manufacturer, or service provider. This connection is built on positive perceptions and actions, such as repeat purchases. Maintaining customer loyalty is vital for businesses, offering advantages such as reduced-price sensitivity among loyal customers, decreased marketing expenses for acquiring new customers, and overall improvements in profitability (Rowley et al. 2005).

Furthermore, research by Raman (1999) and Bowen & Chen (2001) emphasizes that loyal customers become influential marketing forces. They actively endorse and recommend a business, contributing to its positive image and reputation. Additionally, loyal customers, already familiar with products or services, often require less explanation and support, resulting in cost savings for the company (Bowen & Chen, 2001).

2.2 Loyalty Programs (LPs)

2.2.1 Definition of Loyalty Program

A Loyalty Program (LP) is characterized by repeated purchases of specific goods or services over an extended period. These programs, often known as frequent purchase or reward programs, are designed to ensure customer loyalty to a retailer (Shoemaker and Lewis, 1999; Long and Schiffman, 2000; Bell and Lall, 2002; Kopalle et al., 1999; Kim et al., 2001). As per Oliver (1997), a Loyalty Program represents "a deeply held commitment to rebuy or re-patronize a preferred product or service consistently in the future, despite situational influences and marketing efforts that have the potential to cause switching behavior." Dowling & Uncles (1997) and Kumar & Shah (2004) introduce loyalty programs as valuable instruments for retailers, serving as essential tools in the highly competitive retail landscape.

2.2.2 Why to join Loyalty Programs

When considering the value of rewards, there are three significant reasons why consumers are motivated to join loyalty programs. Firstly, the exclusive nature of program rewards, which are accessible only to members, serves as a strong incentive, compelling non-members to participate (Leenheer et al., 2007). Secondly, the frequent reminders and accumulation of reward points not only increase consumers' awareness and interest in the loyalty program but also provide them with a sense of psychological satisfaction in collecting and redeeming these points (Leenheer et al., 2007). Thirdly, loyalty programs reinforce the idea of exclusivity and accentuate the distinction between program members and non-members (Barone and Roy, 2010). Consequently, these programs encourage customers to enroll if they desire to be recognized as part of an exclusive group of privileged clients. Previous research has indicated that loyalty program membership positively influences consumer purchase behavior (e.g., Bolton, Kannan, and Bramlett, 2000; Leenheer et al., 2007; Lui and Yang, 2009). Therefore, it is hypothesized that the favorable impact of loyalty program membership on consumer behavior will also lead to a positive association with sales outcomes.

2.2.3 The role of Rewards

Loyalty programs are thoughtfully designed to show appreciation to customers, considering their purchase history (Yi & Jeon, 2003). These programs provide rewards to boost their attractiveness and acknowledging. Participants in these programs receive incentives and rewards based on their engagement and repeated transactions (Noble & Phillips, 2004). The fundamental assumption is that a rewards program motivates customers to make purchase decisions that consider both their immediate circumstances and the long-term goal of earning a frequent buyer reward, (Gomez et., al 2006, Meyer-Waarden et al., 2007). Put simply, effective loyalty programs guide choices that maximize satisfaction and benefits. This concept aligns with studies by Boulding et al. (1993) and Lemon, White, and Winer (2002), indicating that customers are willing to alter their buying habits to optimize overall value.

These rewards primarily aim to enhance the user experience, foster emotional connections with the brand, and encourage loyalty and repeat business. This mutually

beneficial relationship between customers and businesses is a key element of loyalty programs.

In the case of points accumulation, they are awarded to customers after they make a purchase (Hsee, Yu, Zhang & Zhang, 2003) and according to Kivetz, Urminsky, and Zheng (2006), consumers tend to increase their purchase rate as they approach the required number of points to earn a reward, aiming to reach this goal more rapidly. Consequently, they often remain loyal to a single company. According to Bolton, Kannan, and Bramlett (2000), customers who participate in loyalty programs tend to become loyal due to the perceived benefits they obtain.

It is concluded that rewards increase the purchase behavior of online loyalty program members.

2.2.4 Reward Value and Redemption

The retailer's primary goal is to assess the influence of loyalty programs on customer purchasing behavior, emphasizing that these programs are most effective when they provide substantial reward value and utility. The value, representing the discount on a voucher (Blattberg and Neslin, 1990), plays a crucial role. Higher values increase the likelihood of customer consumption (Bell, Ho, and Tang, 1998), with prior research by Reibstein and Traver (1982) indicating that higher reward values lead to increased voucher usage and, consequently, higher purchase behavior.

O'Brien and Jones (1995) propose that customers evaluate loyalty programs based on two key factors: the value of the rewards offered and the likelihood of redeeming those rewards. Customers assess the perceived worth of rewards in relation to the effort and time required to earn them.

Companies aim to strike a delicate balance between offering attractive rewards and ensuring the easy redemption of accumulated points in loyalty programs. Research by Hsee et al. (2003) and Van Osselaer, Alba, and Manchanda (2004) underscores the psychological impact of loyalty points, enhancing the perceived value of transactions (Thaler, 1985). This contributes to an overall positive perception when engaging with the company.

Understanding the customer perspective is crucial for developing effective loyalty programs that enhance purchases, boost sales, and foster loyalty. Reward value and redemption rate emerge as critical components, crucial for member acquisition and retention. These factors, integral to loyalty program design, contribute to revenue growth and stimulate increased spending among satisfied customers (Humby et al., 2003; Taylor and Neslin, 2005).

The design of rewards significantly influences redemption rates and customer purchase behavior. Elements such as perceived fairness, ease of point accumulation, reward value, and the likelihood of reward attainment impact consumers' attitudes toward scheme membership and redemption (O'Brien and Jones, 1995; Nunes and Drèze, 2006). Sharp and Sharp (1997) emphasize that many individuals grapple with understanding the complexities of points accumulation and redemption.

Consequently, the following hypotheses were derived.

Hypothesis 1: “Reward value has a positive influence towards purchase behavior.”

Hypothesis 2: “Higher reward value increases purchase behavior for customers who use voucher.”

2.3 Age Differences in Loyalty Programs

Younger consumers demonstrate a stronger inclination towards engaging in online purchases compared to their older counterparts, a pattern emphasized by Mathew (2015) who notes the higher frequency of online shopping among the younger demographic. Foscht et al. (2009) propose that establishing early connections with younger customers can deepen relationships and enhance retention opportunities, particularly considering the self-centered and environmentally conscious nature often attributed to this generation.

In contrast, Lian and Yen (2014) identify performance expectations and social influence as primary motivators for older adults engaging in online shopping, while value, risk, and a lack of browsing experience act as significant barriers for this demographic. Unlike older adults, younger online shoppers experience higher motivators and lower barriers in their online purchasing decisions. The observed contrast in online shopping behavior between age groups suggests that older adults may encounter more obstacles in this domain relative to their younger counterparts.

However, concerning loyalty program participation, Patterson (2007) asserts that older customers tend to adopt a more conservative approach to purchasing compared to their younger counterparts. Their resistance to experimenting with new brands or retailers may result in a higher likelihood of demonstrating loyal behavior and increasing purchase behavior.

Regarding rewards, Ward and Davis (1978a) discovered that younger consumers respond more favorably to lower-value vouchers than older consumers. Considering rewards like a price discount offered each time a customer reaches the threshold of ... points, older customers respond more to the level of vouchers and make purchases. For a lower total value of vouchers, younger consumers respond more to their level, while older consumers respond less. With larger amounts of reward values, the model further suggests that the marginal responses to vouchers decrease for the younger consumer group, in contrast to older consumers who become somewhat more responsive.

In this specific scenario, where the value of the rewards is a constant ..€, it's worth noting that some customers may accumulate multiple vouchers and redeem them together in one or more purchases.

Consequently, the following hypothesis was formulated.

Hypothesis 3: "Higher reward value, increases the purchase volume of older customers compared to younger."

2.4 Conceptual Model

The conceptual model, as depicted in Figure 2.4, offers a visual representation of the literature discussed in the previous section and covers the hypotheses outlined in this paper. The model serves as a guide toward realizing this research objective.

To ensure this research's accuracy and reliability, it is important to define how each variable (IV, DV & Moderator) will be measured.

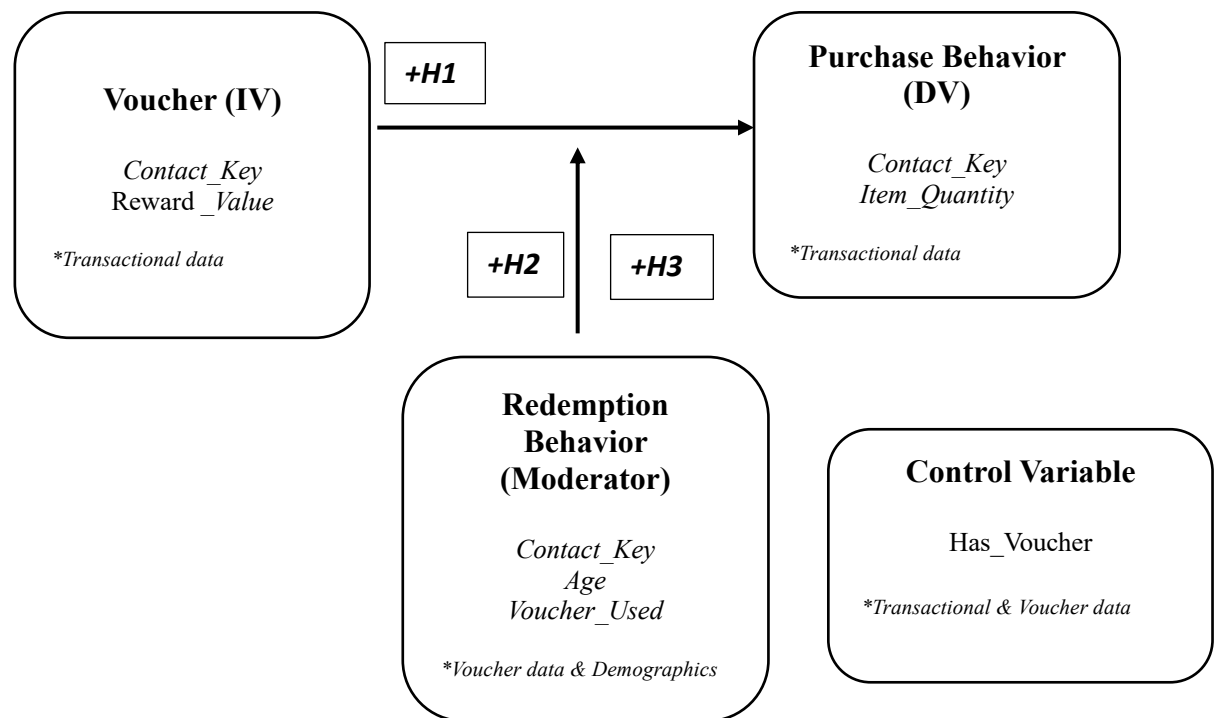


Figure 2.4 Conceptual Model

3. DATA ANALYSIS

In this section of the research, various models and analytical methods are introduced and emphasized to provide a clearer understanding of the techniques employed. This chapter encompasses a discussion of the research approach, the process of gathering data, data preprocessing, and the presentation of descriptive statistics for the variables.

3.1 Research Approach

To assess the impact of voucher ownership on loyalty program members purchasing behavior, we require an extensive dataset. In this context, the main focus is on transactional data from the year 2022, sourced from online members, as this aligns with the voucher dataset's scope. The voucher dataset contains valuable information on the distribution of loyalty vouchers to each member.

To conduct a precise analysis, we will employ a longitudinal approach. This approach allows us to track members' point accumulation patterns, providing valuable insights into their purchase behavior.

3.2 Data Collection

The data for this study is sourced from (company name). It focuses on the transactional behavior of online members in the Dutch market, with a specific emphasis on Netherlands Loyalty Card holders. This dataset spans four years and includes approximately 2 million Loyalty Program Members who have made at least one purchase in the last years. For this analysis, year 2022 will be used.

In addition, the vouchers dataset tracks voucher distribution and redemption for Dutch consumers from 11-01-2021 to 12-04-2023, including voucher start dates ranging from 11-01-2021 to 17-04-2023. Loyalty Program Members accumulate points through purchases, with every ... points (equivalent to ..€ spent) resulting in a ..€ loyalty voucher. These vouchers are automatically sent to members' homes, typically within three weeks, providing flexibility in usage.

3.3 Data Preparation

To facilitate the investigation of how our hypotheses, it is crucial to prepare the dataset for subsequent processing and analysis. This section outlines the process of combining, cleaning, and structuring the datasets. The initial step involves importing the data, ensuring the creation of new variables and the absence of missing values, outliers, or any other errors.

3.3.1 Importing & Cleaning

After importing the dataset into R, the first step involved eliminating empty rows to ensure data cleanliness. Following this, column headers were added to enhance clarity in the subsequent analysis. This procedure was applied to all three datasets: voucher, transactional, and demographics. Additionally, specific variables like "Store Code" and "Pos No" were excluded from the transactional dataset as they were deemed irrelevant for further analysis. Finally, the variables were appropriately transformed into numerical and date formats.

3.3.2 Variable creation

In the initial stages of the data preprocessing, it was aimed to enhance clarity for subsequent analyses. This involved separating variable containing date and time information into distinct variable, focusing specifically on the date of **“Transaction Date”**. Following this, a new binary variable was created, **“Voucher Used”**, derived from the “Voucher Status Code” column. This variable categorizes rows with the term “Used” as 1, indicating that the voucher was utilized, while “Available” or “Canceled” is denoted as 0.

Additionally, with the focus on the transactional & voucher data sets the following steps were conducted:

- The creation of a vector with unique contact keys, derived from the voucher dataset. This vector serves as a reference to identify customers with vouchers. The same approach was used for online22 dataset, to identify customers who are in the voucher dataset and made at least one purchase in online22.

- To determine whether a customer possesses a voucher, we introduced a new column in the transactional dataset called **“Has Voucher”**. This binary column assigns a value of 1 if a customer has a voucher and 0 if they do not. The criterion for this classification is based on the presence of the customer’s “Contact Key” in the vector of unique contact keys.
- To determine whether a customer from the voucher dataset made a purchase, we introduced a new variable **“Purchase”**. The criterion for this classification is based on the presence of the customer’s “Contact Key” in the vector of unique contact keys.
- A novel variable named **“Reward Value”** was generated to represent the value of the reward, specifically the amount of the price discount obtained through points accumulation. This variable was computed as the difference between the original price and the discounted price.

Subsequently, the column **“Age”** from the demographics data set was incorporated.

Finally, it was necessary to merge the data frames into one for the final data frame named online 22.

The datasets were merged and aggregated by the variable unique “Contact Key” and contained the variable which would provide important insights.

*(Variables included: Contact Key, Order Num, Transaction Date, Item Quantity, Original Price, Dis Price, Has Voucher, Voucher Used, Age, Purchase)

3.3.3 Missing values

As outlined by Kang (2013), the presence of missing values can diminish the statistical power of a study and potentially result in inaccurate conclusions due to biased estimations. In our context, the identified missing values in specific variables arose from instances where certain customers did not possess any voucher or did not make any purchase. Therefore, for the purpose of the further analysis, it was opted to exclude these missing values. The focus is on investigating the influence of individuals who possess a voucher, making the removal of these particular missing values pertinent to the scope of our study.

3.3.4 Outliers

Regarding outliers, a careful examination revealed that the variables “Item Quantity,” and “Original Price,” exhibited outlier values. However, these outliers were deemed realistic considering the nature of the data, encompassing the total quantity of items purchased, and the original price per item. So, we kept them like this for the further analysis.

For “Dis Price,” anomalies were identified, particularly instances where values exceeded the original price. One explanation of the anomalies could be that the “Dis Price” is higher because when a customer purchase more than 1 item then the discounted price could be the sum of the purchases, while the original price depicts only the price of 1 item. However, upon closer examination of the data, it was found that this was not the contributing factor.

Furthermore, it was observed that a small percentage (7.6%) of “Dis Price” observations in the data set were higher than the observations of the “Original Price”, so trimming them would not affect the further analysis and the results. Thus, to rectify this issue, extreme values were trimmed, and the data frame replace values in the “Dis Price” column with the corresponding values from the “Original Price” column when the former is greater than the latter. This approach effectively eliminates observations with inflated values, contributing to a more accurate and clarified analysis.

Addressing outliers in the “Age” variable involved a truncation strategy. Outliers were assigned the highest or lowest non-outlier values. If an observation was an outlier on the higher end of the distribution, it was set to the highest value, and vice versa. This approach ensures that extreme values in the “Age” variable do not unduly influence the analysis.

Figure 3.3.4.1 Histogram of Item Quantity.

Figure 3.3.4.2 Histogram of Original Price.

Figure 3.3.4.3 Histogram of Price after discount.

Figure 3.3.4.4 Histogram of Reward Value.

Figure 3.3.5. Histogram of Age.

3.4 Data Description

This section provides detailed explanations of the variables used for analysis. These explanations can be found in Table 3.4. Along with the descriptions, the variable names and example values are provided to enhance clarity and understanding.

Variable	Description	Example Value
Contact Key	A unique identifier of a member	██████
Order Num	A unique identifier of a transaction	██████
Member Key	A unique identifier to identify members	██████
Transaction Date	Date of transaction	██████
Voucher Status Code	Available, Used, Cancelled	██████
Has Voucher	Binary variable, 1= possess voucher, 0= do not possess	█
Voucher Used	Binary variable, 1= used voucher, 0= do not used	█
Item Quantity	Amount of item they purchase	█
Original Price	Original price of the item	████
Dis Price	Price of item after the discount	████
Reward Value	Value of the reward	████
Age	Indicates the age of customers	█

Table 3.4 Variable explanation

3.5 Data Exploration

This section covers the exploratory analysis of the online 22 dataset. Table 3.5 shows the statistical description of the variables of this analysis.

Variables	Min	Median	Mean	Max
Voucher Used				
Item Quantity				
Original Price				
Dis Price				
Reward Value				
Age				

Table 3.5 Descriptive Statistics.

4. METHODOLOGY

To examine and assess the hypotheses, regular Poisson Regression analysis and Truncated Poisson model will be employed. These methods are suitable as the dependent variable, “Item Quantity”, represents a count variable and by definition is larger than 0. The choice of Poisson Regression is justified by the nature of the data, where the mean is expected to be equal to the variance. However, it's worth noting that in this study, an observation of “Underdispersion” was noted, wherein the mean exceeded the variance. Poisson Regression is well-suited to accommodate this specific scenario, as discussed by Haris, Yang, and Hardin in 2012.

4.1 Poisson Regression

A count variable represents discrete values (0, 1, 2, ...) indicating the frequency of an event within a specified time frame. In this context, the dependent variable (Item Quantity) signifies the count of items purchased by a customer from the retailer. Count variables inherently take only non-negative integer values or zero, as it is not possible for a purchase to involve a negative quantity of items.

When dealing with a count variable, the use of Ordinary Least Squares (OLS) is inappropriate due to the assumptions inherent in standard regression analysis. OLS assumes a normal distribution, which is not suitable for count data that can be non-negative, discrete, and exhibit non-continuous behavior. Moreover, it is unrealistic to assume an error term with a normal distribution in this context.

In contrast, the Poisson distribution offers significant advantages over the normal distribution when modeling count data. The Poisson distribution is discrete, assigning probability values exclusively to nonnegative integers. This discrete nature aligns well with the characteristics of count outcomes, where values are naturally limited to integers. The Poisson distribution's probability mass function has the following definition:

$$P(Y_i = Y) = \frac{\lambda^Y \exp(-\lambda)}{Y!} \quad (1).$$

probability that the count variable Y_i takes the value y for subject i .

The underlying assumption of Poisson regression is equidispersion, where the mean and variance are assumed to be equal. So, variance increases automatically as the mean increases. Since we restrict λ to take only positive values, link function is used. So, the Poisson regression model can be depicted as,

$$\ln(\widehat{\mu}) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p \quad (2).$$

Where μ is the predicted count on the outcome variable (Item Quantity) given the specific values on the predictors $X_1, X_2, \dots, X_p$, (Reward Value, Voucher Used, Age). Recall that $\ln$ refers to the natural logarithm, β_0 is the intercept, and β_1 is the regression coefficient for the first predictor X_1 , (Leeftang, et al., 2015).

However, in practical scenarios, equidispersion is not often observed in the data. Typically, the variance exceeds the mean, leading to a phenomenon known as overdispersion (Dean et al., 1992). Conversely, in situations where the variance is smaller than the mean, it is termed underdispersion.

Modeling underdispersed count data using inappropriate models can result in overestimated standard errors and misleading inferences. Although there are a number of methods for dealing with overdispersion in count data, including combinations of Poisson and negative binomial distributions (Yang et al., 2007; Hilbe 2014), there aren't many models for dealing with underdispersed count data. A Poisson regression method was presented by Haris, Yang, and Hardin (2012) with the express purpose of handling situations in which the mean is greater than the variance. However, Poisson model may encounter some violations. In this case it seems that zero events cannot be observed. Despite the fact that the mean is high, the probability to observe 0 events is not exactly 0. For instance, in this study's case the dependent variable "Item Quantity" is by definition larger than 0 (purchased items 1,2,3 etc.).

So, for our investigation we will test Poisson Regression and Truncated Poisson model.

5. RESULTS

In the context of a dynamic and competitive business environment, industries have encountered escalating challenges in recent years. Consequently, the significance of loyalty program rewards has grown substantially for retailers, serving as a pivotal means to convey their mission and prompt meaningful engagement. This section is dedicated to presenting research findings and results related to loyalty program members holding a voucher, with a specific focus on dataset online 22.

Three methods were employed to test the hypotheses:

- **Main Effects Poisson Regression (model 1):** A Poisson regression model was initially fitted, considering only the main effects of the independent variable “Reward Value” and the moderators “Voucher Used” and “Age”. This method aimed to assess how these variables individually influence the dependent variable “Item Quantity” without incorporating interaction terms.
- **Interaction Effects Poisson Regression (model 2):** Subsequently, another Poisson regression model was constructed, but this time the code was modified to include interaction terms between “Voucher Used” and “Age”. These interaction terms were added to examine whether they have a significant impact on the relationship between “Reward Value” and “Item Quantity”.
- **Zero-Truncated Poisson Regression (model 3):** Lastly, a Zero-Truncated Poisson regression was conducted. This type of regression is suitable when the dependent variable is restricted to values higher than zero, as is the case with “Item Quantity”. The model was implemented to account for the non-zero nature of the responses, aligning with the characteristics of the data.

Before delving into the Poisson regression model and Zero Truncated Poisson model, it was imperative to conduct an initial assessment of correlations and multicollinearity among the variables. This preliminary analysis aimed to offer a comprehensive understanding of the interrelationships among the studied variables.

In regression analysis, there exist crucial assumptions regarding the model, such as correlation and multicollinearity. When these assumptions are violated, the reliability

of the model diminishes, rendering it unsuitable for accurately estimating population parameters (Daoud, J. I., 2017).

5.1 Correlation

The correlation analysis in Table 5.1.1 and Figure 5.1.1 reveals patterns among key variables. A moderate negative correlation between “Item Quantity” and “Reward Value” suggests that as the price value of the reward increases, there is a tendency for a decrease in the quantity of items purchased. Meanwhile, minimal correlations are observed between “Item Quantity” and both “Voucher Used” and “Age”, indicating weak linear relationships. Voucher usage and the age of customers do not strongly correlate with the quantity of items purchases. Additionally, a small negative correlation between “Voucher Used” and “Reward Value” hints at a weak tendency for lower prices when vouchers are used, but the correlation is not strong. Similarly, a minimal correlation is noted between “Age” and “Reward Value”, suggesting a weak linear relationship between the price value of rewards and the age of customers. Lastly, a small positive correlation between “Voucher Used” and “Age” implies a weak tendency for voucher usage to increase with customers age, though the correlation is not strong enough.

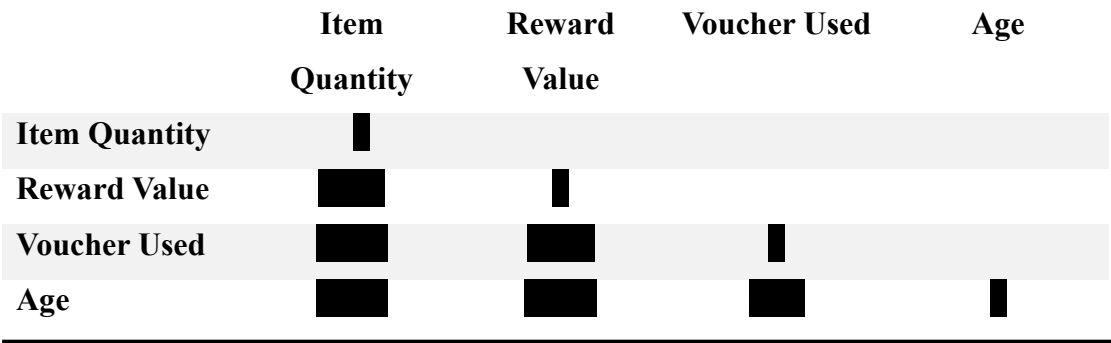


Table 5.1 Correlation Matrix.

Figure 5.1 Correlation Plot.

5.2 Multicollinearity

A statistical phenomenon known as multicollinearity occurs when there is strong connection between two or more predictor variables in a regression model. Although multicollinearity to some extent may not cause major problems, its presence, especially at moderate to high levels, can cause challenges with the regression analysis (Daoud, J. I., 2017).

Relying solely on pairwise correlations between predictors has limitations, as the perception of a correlation's magnitude can be subjective and contingent on individual perspectives or the specific field of research. To address this, the use of variance inflation factors (VIF) becomes essential. VIF is a valuable indicator that helps quantify the extent to which the variance of predictor coefficients is inflated when multicollinearity is present.

All VIF values, as shown in Table 5.2, are roughly 1, indicating that there isn't any serious multicollinearity among the model's predictors. This finding is encouraging, indicating that the predictors do not exhibit high correlation with each other. The VIF analysis provides a more robust measure to assess multicollinearity, considering the impact on standard errors and variance inflation of predictor coefficients.

	Voucher Used	Age	Reward Value
Item Quantity			

Table 5.2 Multicollinearity.

5.3 Testing Hypotheses

5.3.1 Poisson Regression of Main Effects Model (model 1)

The analysis revealed that the intercept and the variables “Reward Value”, and “Age”, exhibited statistical significance, with p-values below the common threshold of 0.05. Conversely, “Voucher Used” lacked statistical significance, as indicated by p-values exceeding 0.05, suggesting a lack of compelling evidence for their influence.

Conducting the exponential of the significant variables, the following results emerged:

- **Intercept:** The exponentiated intercept is [REDACTED], suggesting that when all other predictors are zero, the expected count of “Item Quantity” is [REDACTED] times higher.
- **Reward Value:** The exponentiated coefficient [REDACTED] indicating that, holding other variables constant, a one-unit increase in “Reward Value” is associated with a [REDACTED] decrease in the expected count of “Item Quantity”.
- **Voucher Used:** Not statistically significant, it does not have a significant impact on the expected count of “Item Quantity”.
- **Age:** The exponentiated coefficient [REDACTED] suggesting that, holding other variables constant, a one-unit increase in “Age” is associated with a very small [REDACTED] in the expected count of “Item Quantity”.

Variables	Estimates	Std. Error	P-value	Exponential
Intercept	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]
Reward Value	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]
Voucher Used	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]
Age	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]

Significance codes: 0 ‘***’ 0.011 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘’ 1

Table 5.3.1 Summary of Poisson Regression Model.

5.3.2. Poisson Regression of Interaction Effects Model (model 2)

The analysis revealed that the intercept and the variables “Reward Value”, “Age”, and the interaction term “Reward Value: Age” exhibited statistical significance, with p-values [REDACTED] the common threshold of [REDACTED]. Conversely, “Voucher Used” and “Reward Value: Voucher Used” lacked statistical significance, as indicated by p-values [REDACTED], suggesting a [REDACTED] evidence for their influence.

Conducting the exponential of the significant variables, the following results emerged:

- [REDACTED]: The exponentiated intercept is [REDACTED], suggesting that when all other predictors are zero, the expected count of “Item Quantity” is [REDACTED] times higher.

- **Intercept:** The exponentiated coefficient is 0.000000, indicating that, holding other variables constant, a one-unit increase in “Reward Value” is associated with a 0.000000 decrease in the expected count of “Item Quantity”.
- **Voucher Used:** 0.000000 impact on the expected count of “Item Quantity”.
- **Age:** The exponentiated coefficient is 1.000000, suggesting that, holding other variables constant, a one-unit increase in “Age” is associated with a 1.000000 increase in the expected count of “Item Quantity”.
- **Reward Value: Voucher Used:** 0.000000 impact on the expected count of “Item Quantity”.
- **Reward Value: Age:** The exponential coefficient is 0.999999. This indicates that a one-unit increase in the interaction term is associated with a minimal decrease of 0.000001 in the expected count of “Item Quantity”. It's important to note that the effect size is minimal, indicating that the interaction between “Reward Value” and “Age” has only a slight impact on the expected count of “Item Quantity”.

Variables	Estimates	Std. Error	P-value	Exponential
Intercept	0.000000	0.000000	0.000000	0.000000
Reward Value	0.000000	0.000000	0.000000	0.000000
Voucher Used	0.000000	0.000000	0.000000	0.000000
Age	1.000000	0.000000	0.000000	1.000000
Reward Value: Voucher Used	0.000000	0.000000	0.000000	0.000000
Reward Value: Age	0.999999	0.000000	0.000000	0.999999

Significance codes: 0'***' 0.011'***' 0.01'*' 0.05'.' 0.1'' 1

Table 5.3.2 Summary of Poisson Regression Model with Interaction Terms.

5.3.3. Zero-Truncated Poisson Regression (model 3)

The analysis revealed that almost all the variables of the model exhibited **0.000000** with p-values **0.000000** the common threshold of 0.05, **0.000000** the variables **0.000000**.

Conducting the exponential of the significant variables, the following results emerged:

- **Intercept 1 (Truncated negbin model):** The exponential of the [REDACTED] for the log mean is [REDACTED] suggesting that when all other predictors are zero, the expected count of “[REDACTED]” is [REDACTED] times [REDACTED]
- **Intercept 2 (Dispersion parameter):** The negative binomial distribution’s size is approximately [REDACTED]
- **Reward Value:** The exponentiated coefficient is [REDACTED] indicating that, holding other variables constant, a one-unit increase in [REDACTED] is associated with a [REDACTED] in the expected count of “[REDACTED]”
- **Voucher Used:** [REDACTED], it [REDACTED] impact on the expected count [REDACTED]
- **Age:** The exponentiated coefficient is [REDACTED], suggesting that, holding other variables constant, a one-unit increase in [REDACTED] is associated with a [REDACTED] [REDACTED] in the expected count of [REDACTED]
- **Reward Value: Voucher Used:** [REDACTED] it [REDACTED] [REDACTED] impact on the expected count of “[REDACTED]”
- **Reward Value: Age:** The exponential coefficient is [REDACTED]. This indicates that a one-unit increase in the interaction term is associated with a minimal decrease of [REDACTED] in the expected count of [REDACTED]. It's important to note that the effect size is [REDACTED] indicating that the interaction [REDACTED] [REDACTED] has only a [REDACTED] on the expected count of [REDACTED]

Variables	Estimates	Std. Error	P-value	Exponential
Intercept:1	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]
Intercept:2	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]
Reward Value	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]
Voucher Used	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]
Age	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]
Reward Value: Voucher Used	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]
Reward Value: Age	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]

Significance codes: 0‘***’ 0.011‘**’ 0.01‘*’ 0.05‘.’ 0.1‘’ 1

Table 5.3.3 Summary of Zero Truncated Poisson Model.

5.4 Model fit & Model selection

Several measures of goodness-of-fit for the Poisson regression model are available in the literature, with the Akaike information criterion (AIC) and Bayesian Information Criterion (BIC) being commonly utilized (Gagné & Dayton, 2022).

The core principle of the AIC involves penalizing the introduction of extra variables into a model, imposing a penalty that increases the error when incorporating additional terms. The formula AIC is given by $AIC = 2k - 2\log(L)$, where L represents the likelihood of the model given the data, and k is the number of parameters. The BIC serves as a refinement of the AIC, implementing a more pronounced penalty for the inclusion of supplementary variables in the model (Brewer et al., 2016). Its calculation is expressed as $BIC = k \log(n) - 2 \log(L)$, where n is the sample size. Lower AIC and BIC values signify a better-performing model as they penalize the addition of unnecessary variables (Kuha et al., 2014).

Based on the outcomes of the likelihood ratio test, AIC suggests that [redacted] fits better, while BIC indicates [redacted]. Consequently, for Hypothesis 1, [redacted] is chosen to explore how [redacted]” influences the [redacted]” without [redacted]. However, for Hypotheses 2 and 3, [redacted] is selected to investigate how “[redacted]” considering the [redacted]

	AIC	BIC
Null		
Model 1		
Model 2		
Model 3		

Table 5.4 Summary of AIC / BIC.

6. DISCUSSION

6.1 General Discussion

In this chapter, the primary objective is to evaluate the effectiveness of quantity items purchased by customers who possess a voucher. To support this research goal, Table 6.1 presents a comprehensive summary of hypothesis testing. The table encapsulates the results, assessing how well the collected data align with or deviate from the research hypotheses. Considering the model fit criterion, for testing Hypothesis 1, we opted for model 1, aiming to examine the singular impact of “Reward Value” on the dependent variable “Item Quantity” without introducing interaction effects. Conversely, for Hypotheses 2 and 3, we chose model 2, as the objective was to explore how “Reward Value” influences “Item Quantity” while considering the potential interaction effects of “Voucher Used” and “Age”.

Hypothesis	Assumption	Support	Findings
H1	“Reward value positively influence the purchase behavior”	Rejected	
H2	“Higher value of the reward increases purchase behavior for voucher users”	Rejected	
H3	“Higher value of reward, increases the purchase behavior of older compared to younger”	Rejected	

Table 6.1 Summary of Hypotheses Testing.

Hypothesis 1: suggests that reward value impact positively the purchase behavior of loyalty program members. This proposition aligns with previous research findings by Bell, Ho, and Tang (1998), suggesting that a larger reward value enhances the likelihood

of customers making purchases. Furthermore, it is consistent with the idea that loyalty programs are most impactful when they provide substantial rewards, as emphasized by Blattberg and Neslin (1990). This strategic approach has been associated with heightened spending among satisfied customers, as highlighted in studies by Humby et al. (2003) and Taylor and Neslin (2005).

However, the results of Poisson regression with main effects (model 1) led to the rejection of the hypothesis. This decision is based on the presence of a negative coefficient, suggesting that as the reward value increases by one-unit, the count of item quantity decreases by [REDACTED]. One contributing factor to this outcome could be the delay in voucher distribution, which might also impact customers' purchasing behavior. Issues such as customer complaints or forgetting about the voucher could potentially influence the observed results.

Hypothesis 2: suggests that a higher value of rewards enhances the purchase volume for loyalty program members who used voucher. This aligns with earlier studies by Reibstein and Traver (1982), which suggested a positive association between larger reward amounts and heightened voucher usage. Following this, O'Brien and Jones (1995), along with Nunes and Drèze (2006), asserted that the perceived value of rewards significantly influences consumers' attitudes toward redemption and purchase behavior.

However, the results of the Poisson regression with interaction effects (model 2) led to the rejection of Hypothesis 2. This decision is based on the lack of statistical significance in the interaction term between the variables "Reward Value", and "Voucher Used", indicating insufficient evidence to support the hypothesis.

Hypothesis 3: posits that a higher value of rewards would increase the purchase volume, particularly the number of items purchased by older customers. This aligns with the notion that younger consumers respond more favorably to lower-value vouchers, while higher-value vouchers lead to diminishing marginal responses in the younger consumer group. In contrast, older consumers tend to show increased responsiveness to higher-value vouchers (Ward and Davis, 1978a).

However, the outcome of the Poisson regression with interaction effects (model 2) has resulted in the rejection of Hypothesis 3. This decision is based on the presence of a negative coefficient, suggesting that as the age of customers and the reward value increase by one unit, the count of items purchased decreases by [REDACTED]. This problem may stem from the fact that older individuals, despite being more inclined towards valuable vouchers, encounter challenges in redeeming them, such as difficulties with internet usage (Solomon et al., 2010). This barrier might therefore have an effect on their purchase decisions.

6.2 Managerial Implications

The analyses conducted in this study unveil challenges and opportunities for managers and businesses to optimize their loyalty program rewards. The research considered one crucial factor for the analysis the value of the reward, and the moderator factor voucher usage and customers' age. Three key recommendations are put forth for the managerial board based on the hypothesis's outcomes.

- **Lower or Personalized Rewards (H1):** In contrast to what was first thought, the study indicates that, higher reward values are linked to a decline in customer buying behavior. The recommendation for managers is to consider offering lower-value rewards. This strategic approach is deemed advantageous not only from a cost-effectiveness perspective but also aligns with the empirical findings of this study. Another suggestion is to provide specific reward regarding customers preferences. Finding the ideal incentive value that appeals to the target audience can be assisted by carrying out in-depth market research and comprehending consumer preferences.
- **Consideration for customers who do not use vouchers (H2):** Given the inconclusive evidence regarding increased purchase behavior among voucher utilization segments, managers may explore offering higher reward values specifically to customers who have not utilized vouchers. By closely monitoring and analyzing the reactions of these distinct segments, managers can make data-driven decisions to tailor rewards to specific customer behaviors, fostering a more effective strategy.

- **Targeting Younger Segments for Vouchers (H3):** Results of the Poisson regression show that younger customers exhibit stronger connections to higher values rewards. Therefore, when implementing voucher-based loyalty programs, managers should consider targeting younger customers instead of older as was stated in the hypothesis. This approach aligns with the results and their preferences and may lead to increased engagement within business.

6.3 Conclusion

In conclusion, this thesis thoroughly explores the intricate dynamics of rewards within loyalty programs (LPs), with a specific focus on the role of points accumulation. The research begins by highlighting the crucial role of loyalty programs in fostering enduring connections between customers and businesses, renowned for their potential to drive revenues, enhance profits, and increase customer retention, especially within valuable customer segments (Bolton, Kannan, and Bramlett, 2000).

Thus, the following research questions was derived:

“To what extent does the possession of Points Accumulation influence the purchase behavior of online loyalty program members?”

This study examines the influence of a crucial factor, “Reward Value”, along with the interaction terms “Voucher Used” and “Age”, on the purchase volume (“Item Quantity”) of loyalty program members. The analysis involved three consecutive Poisson regressions: one focusing on main effects (model 1), another on interaction effects (model 2), and the last utilizing a zero-truncated Poisson model (model 3). However, based on the model fit criterion, models 1 and 2 were chosen for further exploration of Hypotheses 1 and 2 & 3, respectively.

The initial hypothesis posited that the positive impact of reward value on the purchase intention of loyalty program members is supported by Bell, Ho, and Tang (1998), who asserted that higher reward value enhances the probability of customers making a purchase. Contrary to the initial expectation, the outcome of the Poisson regression with

main effects (model 1) revealed an inverse relationship. Specifically, higher reward values were associated with a decrease in item quantity. A factor that may contribute to this outcome is the delay in the distribution of vouchers, which might also play a role in customers' purchase behavior. Customer complaints or forgetting about the voucher could be potential factors influencing the observed results.

Hypothesis 2 delves into the impact of reward value on the purchasing behavior of customers who have utilized vouchers. Building on O'Brien and Jones' (1995) proposition that customers assess loyalty programs based on reward value and drawing on prior research by Reibstein and Traver (1982) suggesting that higher reward values encourage increased voucher usage.

However, the findings present a lack of compelling evidence to support Hypothesis 2. This discrepancy emerged during the Poisson regression analysis, where the interaction term between voucher usage and reward value did not achieve statistical significance, with a p-value exceeding the common baseline of 0.05.

The third hypothesis initially introduces age as a moderator, acknowledging the diverse dynamics of customer loyalty across different age groups. Ward and Davis (1978a) observed in their study that younger consumers respond more strongly to lower-value vouchers, while older consumers tend to become somewhat more responsive to high-value vouchers. Despite the individual positive influence of "Age" on the expected count of "Item Quantity", the primary focus of the hypothesis is on the interaction term of "Reward Value" and "Age", revealing a contradictory finding. Specifically, higher reward value decreases the purchase behavior of older customers.

This inconsistency may stem from the possibility that older individuals, while being more attracted to high-value vouchers, encounter challenges in applying them, resulting in a lack of impact on their purchase behavior. Consequently, prioritizing older customer segments may not be advantageous for retailers when considering the value of rewards.

In conclusion, this thesis presents valuable insights into the intricate dynamics of loyalty program rewards. By scrutinizing the key factors discussed above, businesses and managers can extract actionable insights to optimize their loyalty programs and improve customer purchase behavior.

The suggestion is for managers to consider offering lower-value rewards or personalized rewards tailored to customers' preferences, which not only proves advantageous due to cost considerations but also aligns with the findings of this study. Additionally, when offering vouchers, targeting younger segments is recommended for optimal results.

However, when it comes to customers who have used vouchers, the evidence is inconclusive regarding an increase in their purchase behavior. To address this, one potential solution could involve offering higher reward values to customers who have not utilized vouchers, saw their reaction and compared them with customers who utilized their voucher.

Overall, the research underscores the necessity for a nuanced and context-specific approach to loyalty program design, taking into account the diverse factors influencing customer behavior in the ever-evolving landscape of business-consumer relations.

6.4 Limitations & Future Research

While this research makes valuable academic contributions, it is essential to acknowledge and address potential limitations that could impact the reliability of the findings. In this section, we delve into these limitations and provide detailed recommendations.

A notable limitation revolves around loyalty programs, as emphasized by Rothschild & Gaidis (1981) and Uncles et al. (2003). When loyalty programs offer attractive rewards, customers might form strong bonds with the program itself rather than with a specific brand. Importantly, removing the reward factor can diminish the positive effects of the loyalty program, potentially affecting repeat purchases. Future research could explore the use of consumer data to tailor rewards to individual interests and purchasing histories, adding a personal touch to products and strengthening the consumer-brand relationship. This holistic approach would enhance both the structural appeal of the loyalty program and individualized, meaningful interactions with customers.

An additional noteworthy limitation involves delays or challenges in redeeming rewards, as highlighted by Dowling and Uncles (1997) and Zhang et al. (2000). These obstacles have the potential to reduce the effectiveness of the loyalty program in encouraging desired purchase behavior and nurturing customer loyalty. The timing of voucher delivery varies, with some customers receiving vouchers promptly and other facing delays.

Therefore, future research focusing on specific timeframes and days associated with receiving and redeeming vouchers would be more pertinent, offering valuable insights for managers seeking to understand customer purchase behavior.

Additionally, the study's findings are derived from a specific sample, potentially limiting the generalizability of the results. The research concentrated on company name within the Netherlands, and as such, the outcomes may not be the same to other European markets. Researchers should encourage to undertake analogous studies across various regions and with diverse samples to improve the generalizability of the findings.

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